Fiscal Year:	FY 2012	**Task Last Updated:**	FY 03/27/2012
PI Name:	Held, Kathryn Ph.D.		
Project Title:	Mechanisms for Induction of Bystander Effects by High Energy Particles in Cells and Tissues		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	kheld@partners.org	**Fax:**	FY 617-724-8320
PI Organization Type:	NON-PROFIT	**Phone:**	617-726-8161
Organization Name:	Massachusettes General Hospital		
PI Address 1:	Radiation Oncology		
PI Address 2:	55 Fruit Street		
PI Web Page:			
City:	Boston	**State:**	MA
Zip Code:	02114-2696	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2006 Radiation Biology NNJ06ZSA001N
Start Date:	07/01/2007	**End Date:**	12/31/2011
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
Contact Email:	Yuri.V.Griko@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: NCE to 12/31/2011 awarded (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
Grant/Contract No.:	NNX07AE40G		
Performance Goal No.:			
Performance Goal Text:			
	In space, astronauts are exposed to a complex and unique radiation environment composed of trapped proton and electron radiation, galactic cosmic rays (GCR; energetic protons and heavy charged ions, HZE), and solar particle events (SPE), primarily protons. Although these radiations can result in appreciable cumulative doses to astronauts during long-duration missions, the exposures occur at low fluences and fluence rates, such that particle traversals through individual cells in an astronaut's body may be well separated in tissue location and time. Under such conditions, so-called non-targeted effects such as bystander responses may have increased significance. The radiation-induced bystander effect is the occurrence of biological responses in unirradiated cells near to or sharing medium with cells actually traversed by radiation. We hypothesize that at the low fluences of protons and HZE particles to which astronauts are exposed in space, a significant fraction of the cellular changes may be due to bystander responses, in		

Task Description:	addition to the damages caused in cells directly traversed by ionizing particles or their associated delta rays. We have the following specific aims. 1. Test the hypothesis that bystander effects occur at particle fluences so low that only 1-2 % of cells in a culture dish are traversed by an energetic proton or HZE particle. 2. Extend our low fluence work to studies using repeated exposures of cells to low fluences and mixed (actually sequential) beams of more than one particle type. This will test the hypothesis that cells that have survived an initial traversal and sent a bystander signal, can again send a signal when traversed once again by a particle or can subsequently respond as bystanders to a hit in a neighboring cell. 3. Our studies have indicated that reactive oxygen species (ROS) are at least partially involved in transmittal of the Fe ion-induced bystander signal from hit to unirradiated cells. We will test the hypothesis that non-nuclear damage-initiated, metabolically produced ROS in hit cells are involved in the early steps of bystander signaling. Additionally, we will investigate chemical countermeasures to reduce the bystander responses based on removal of ROS or prevention of their formation. 4. In tissues, cells do not exist in isolation, and there is increasing interest in how signaling between cell types (especially stromal and epithelial) influences cellular responses to stresses. We will test the hypothesis that radiation-induced bystander signals can be sent between differing cell types, but the characteristics of the responses depend on the cell types and the types of radiation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Although our studies are specifically geared to providing information on the possible biological effects, especially cancer induction, of low fluences of energetic particles on astronauts, the data may have relevance to cancer therapy and the effects of low doses of radiation on radiation workers (e.g., those involved in clean-up of radioactive sites) and the general public. With regards to cancer, there is increasing use of protons (energies of 65-250 MeV) as an effective treatment modality in radiation oncology because they provide a great therapeutic advantage in some types of cancers due to exquisite dose localization. Over 60,000 patients have been treated world-wide with protons, and several new proton centers have opened in the last few years or are under construction in the US. However, despite the clinical success of protons, some concerns exist regarding the possible side-effects, including the possibility of cancer caused by the low doses received by normal tissues. Additionally, there is increasing interest in the use of carbon ions in cancer therapy, with carbon ion treatment centers now in use in Japan and Germany, and the potential being discussed in the US. Our studies could provide important and useful new insight on the effects of these energetic particles on biological systems and the mechanisms by which they act. Such findings would be expected to translate into helping to develop effective ways to reduce possible side-effects, including cancer, from the treatments.
Task Progress:	This report covers a six-month, no-cost-extension (NCE) period of the grant, with limited budget, during which we had two goals: (1) continue to count and analyze micronuclei (MN) and 53BP1 foci samples from irradiations at NSRL for the study of the LET and/or track structure dependence of the minimum particle fluence needed for increasing damage in irradiated and bystander cell populations exposed to very low fluences of charged particles (extension of aim 1); (2) continue studies with mixed cell cultures and 3D epithelial tissue model (aim 4). During this NCE we published a paper (Yang et al. 2011) showing that at very low iron ion (1 GeV/n) fluences, i.e., 2-50% of cells traversed by an ion, or very low proton (1 GeV) fluences, i.e., 2-200 protons/cell, there was increased damage in cells in both irradiated and bystander populations. That finding--that about 100-fold lower particle fluence is needed to produce DNA damage in irradiated or bystander populations with Fe ions than with protons of the same energy--suggested that linear energy transfer (LET) or particle track structure, including delta-rays, is a determinant of the minimum dose needed for the effect. Our studies with 300 MeV/n Fe ions (higher LET) show a shift to even lower minimum particle fluence for induction of the effect, consistent with LET being important. Ongoing analysis of samples irradiated with silicon and carbon, of intermediate LET, appear to also be yielding data consistent with that idea. We expect to complete these studies soon and submit a manuscript for publication. The studies with mixed cell types (fibroblasts and keratinocytes) in 2D and with the same cells in a 3D model of epithelial tissue (aim 4) have been completed. The findings have been written up in a PhD thesis (Lumpkins, 2012), and two manuscripts are being prepared. In summary the data in monolayers show that with either X-rays or protons (low LET radiations), there is a 1.5-2-fold increase in MN formation in bystander fibroblasts or keratinocytes after either cell type is irradiated, but there is no significant increase in MN in bystander fibroblasts in co-culture with irradiated keratinocytes. In the 3D studies, interestingly, the data in irradiated cells show that the fraction of fibroblasts with DNA damage when irradiated in the 3D constructs is significantly less (4-fold at the shortest time of 1 h) than in cells irradiated in 2D monolayers. With time after irradiation, the fraction of damaged cells decreases such that at 24 h it is back to background in cells in 3D, but remains elevated out to 72 h in cells in 2D. In contrast to the irradiated cells in 3D, the level of damage in the bystander portion of the 3D model is low at 1 h, but increases to a peak at 12 h after protons and a maximum at 24-48 h after Fe ion exposure. Importantly, the data show that bystander signals can travel considerable distances (mm), consistent with the work of others. Lastly, the data show that, although scavengers have no effect on DNA damage in irradiated cells in 3D, the level of DNA damage in bystander cells in 3D is reduced nearly to control, unirradiated, levels by scavenging hydrogen peroxide, superoxide and NO in the medium shared by the construct halves, findings similar to those in 2D. In studies to assess intercellular signaling between normal fibroblasts and tumor cells, we have shown that there is no bystander signaling with a human chrondrosarcoma cell line (HTB94) irradiated with X-rays, protons or iron ions. However, irradiated HTB94 cells can send signals to bystander fibroblasts, to elicit DNA damage, although the tumor cells do not respond to signals from the fibroblasts. A paper describing these data is in press (Wakatsuki et al., 2012).
Bibliography Type:	Description: (Last Updated: 11/13/2018)

Abstracts for Journals and Proceedings	Held KD, Yang H, Magpayo N. "LET dependence of effects of very low fluences of charged particles on irradiated and bystander cells." Poster presented at the 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. 14th International Congress of Radiation Research, Warsaw, Poland, August 28-September 1, 2011. Abstract Book, p. 156. , Aug-2011
Abstracts for Journals and Proceedings	Yang H, Magpayo N, Held KD. "Further Studies on the Effects of Very Low Fluences of Charged Particles on Irradiated and Bystander Cells." Presented at the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. Abstracts of the 22nd Annual Space Radiation Investigators' Workshop, League City, TX, September 18-21, 2011. http://www.dsls.usra.edu/meetings/radiation2011/pdf/7021.pdf , Sep-2011
Abstracts for Journals and Proceedings	Wakatsuki M, Magpayo N, Held KD. "LET-dependent differences in bystander stimulation induced by X-rays and charged particle irradiation in chondrosarcoma cells." Poster presented at 53rd Annual Meeting of ASTRO, Miami, FL, October 2-6, 2011. Abstracts of the 53rd Annual Meeting of ASTRO, Miami, FL, October 2011. , Oct-2011
Articles in Peer-reviewed Journals	Yang H, Magpayo N, Rusek A, Chiang I-H, Sivertz M, Held KD. "Effects of very low fluences of high energy protons or iron ions on irradiated and bystander cells." Radiat Res. 2011 Dec;176(6):695-705. Epub 2011 Oct 12. PMID: 21988573 , Dec-2011
Articles in Peer-reviewed Journals	Wakatsuki M, Magpayo N, Kawamura H, Held KD. "The lack of a bystander response induced by X-rays, protons and heavy ions in chondrosarcoma cells." Int J Radiat Oncol Biol Phys. In press. February 2012. , Feb-2012
Articles in Peer-reviewed Journals	Lumpkins S, Anzenberg V, Yang H, Magpayo N, Held KD. "Radiation-induced intercellular signaling between human keratinocytes and fibroblasts." Radiat Res. In preparation, March 2012. , Mar-2012
Articles in Peer-reviewed Journals	Lumpkins S, Rajadurai A., Yang H, Magpayo N, Held KD. "Space radiation-induced bystander signaling in a 3D skin tissue model." Radiat Res. In preparation, March 2012. , Mar-2012
Dissertations and Theses	Lumpkins S. "Space radiation-induced bystander signaling in 2D and 3D skin tissue models." Ph.D. Thesis, Massachusetts Institute of Technology, Cambridge, MA, February, 2012. , Feb-2012